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THERMAL RUNAWAY DETECTION AND COTAINMENT MECHANISM

Abstract

A thermal runaway protection system for a battery can include a divider wall, a protection structure, and a piston. The divider wall can be adjacent to a battery module enclosure. The protection structure can be attached to the battery module enclosure. The piston can be disposed in the protection structure, where a first piston end is in fluid communication with an enclosure interior space of the battery module enclosure via a pressure hole, and where a second piston end is proximate to or attached to the divider wall. The system can include a spring pushing the piston towards an untriggered position to control a triggering pressure. The system can include a set of lock pawls to retain a triggered position of the piston during a thermal runaway event while creating an air gap between the battery module enclosure and the divider wall.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to a system of thermal runaway detection and containment for a battery, and in particular to a system for passive cooling mechanisms for thermal runaway protection for a group of battery enclosures.

BACKGROUND

[0002] Generally, electric vehicles, mobility machines, aircraft, and devices that use battery packs or sets of battery cells stacked or grouped together in a battery case or housing need cooling structures and systems to manage the heat dissipated while batteries charge and discharge. Having a large quantity of battery cells stacked or grouped together in a same battery case or housing can compound heat dissipation and cooling issues.

[0003] One disadvantage of using batteries as an electrical power source is that during heavy use or loading of the electrical systems, such as electrical motors driving the propulsion and movement of an automobile or aircraft, the large currents and rapid discharges of the battery cells can lead to rapid and cascading heat buildup within the battery case or housing. Such rapid and cascading heat buildups can result when the heat builds up faster than the heat is being dissipated. This can lead to thermal runaway, especially in lithium-ion batteries, which are commonly used for many electric vehicle systems. When thermal runaway occurs, heat generated by chemical reactions in the battery cells can become self-sustaining and rapidly increase, which can cause a release of flammable substances and gases leading to an explosion or fire.

[0004] There are many potential causes of thermal runaway. Thermal runaway can result from overcharging a battery pack beyond its design parameters, such as during rapid charging using electrical power higher than specified for a battery system. Thermal runaway can result from overdischarging a battery pack, such as discharging using high discharge current beyond its design parameter causing a breakdown of internal components of a battery cell. Thermal runaway can result from operating electrical devices, such as electrical motors in a very hot environment, such as on a very sunny day in the summer without a battery thermal management system. Thermal runaway can also result from physical damage to a battery structure or manufacturing defects, such as the presence of impurities.

[0005] Lithium-ion batteries may be the best battery solution regarding applications that require an efficient and lightweight electric energy storage system. However, lithium-ion batteries may lead to catastrophic failure if a propulsion battery is not designed or handled properly. Lithium-ion batteries may fail due to various reasons, such as, overcharge, over-discharge, overtemperature, etc. The failure of a lithium-ion battery may cause overheating and may potentially lead to explosion. To mitigate such safety hazard, propulsion batteries may incorporate additional active and passive protection features. And because the propulsion battery may need to be certified before the commercialization, safety measures and safety systems may need to be implemented to pass certification tests.

SUMMARY

[0006] An embodiment thermal runaway protection system for a battery includes a battery module enclosure, a divider wall, a protection structure, and a piston. In the embodiment, the battery module enclosure bounds an enclosure interior space. In the embodiment, the divider wall is adjacent to the battery module enclosure. In the embodiment, the protection structure is attached to the battery module enclosure. In the embodiment, the protection structure includes a first chamber inner surface proximate to a first structure end, where the first chamber inner surface at least partially bounds a first chamber interior space inside the protection structure, where the protection structure has a pressure hole at the first structure end, where the pressure hole is in fluid

communication with the enclosure interior space of the battery module enclosure, where the protection structure has a vent hole through a second structure end, and where the vent hole is in fluid communication with the first chamber interior space. In the embodiment, the piston is at least partially disposed in the protection structure, where the piston at least partially separates the enclosure interior space from the first chamber interior space, where the piston has a first piston end and a second piston end, where the piston fits in, and slidably engages with, the first chamber inner surface, where at least part of the first piston end is in fluid communication with the enclosure interior space via the pressure hole, and where the second piston end is proximate to the divider wall.

[0007] A battery system embodiment includes a battery case, a first battery module enclosure, a second battery module enclosure, a divider wall, a first protection structure, and a first piston. In the embodiment, the first battery module enclosure is disposed in the battery case, where the first battery module enclosure bounds a first enclosure interior space. In the embodiment, the second battery module enclosure is disposed in the battery case. In the embodiment, the divider wall is disposed in the battery case, where the divider wall is between the first battery module enclosure and the second battery module enclosure. In the embodiment, the first protection structure has a first pressure hole. In the embodiment, the first piston is at least partially disposed in the first protection structure, where the first piston fits in, and slidably engages with, the first protection structure, where at least part of a first piston end of the first piston is in fluid communication with the first enclosure interior space via the first pressure hole, and where the first piston is configured to move, in response to a pressure above a selected threshold pressure inside the first enclosure interior space acting upon the first piston end, in a first direction in the first protection structure and to separate the first battery module enclosure from the second battery module enclosure to form a separation air gap disposed between the first battery module enclosure and the second battery module enclosure.

[0008] A method embodiment of assembling a thermal runaway protection system for a battery includes providing a battery module enclosure including an enclosure interior space, providing a divider wall, providing a protection structure including a first chamber inner surface proximate to a first structure end, where the first chamber inner surface partially bounds a first chamber interior space, where the protection structure further includes a pressure hole at the first structure end, where the protection structure further includes a set of vent holes through a second structure end, where the vent holes are in fluid communication with the first chamber interior space, and where the protection structure further includes a set of lock pawl pockets in the first chamber inner surface, providing a piston including a first piston outer surface at a first piston end, where the piston further includes a second piston outer surface at a second piston end, where the first piston end has a first piston end width, where the second piston end has a second piston end width, where the first piston end width is greater than the second piston end width, placing a set of lock pawls in the set of lock pawl pockets, placing a spring in the first chamber interior space inside the protection structure, placing the piston into the protection structure such that the spring is surrounding the piston, such that the first piston end fits in and slidably couples with the first chamber inner surface, such that the first piston outer surface contacts the first chamber inner surface, such that the first piston end is in the first chamber interior space, such that the spring is configured to bias between the first piston end and the second structure end, and such that at least part of the first piston end is in fluid communication with the enclosure interior space via the pressure hole, attaching the second structure end to the battery module enclosure, such that the pressure hole is in fluid communication with the enclosure interior space of the battery module enclosure, and attaching the second piston end to the divider wall, such that the divider wall is proximate to the battery module enclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The above and other features and advantages of the present disclosure can be more apparent based on the following detailed description taken in conjunction with the accompanying drawings, in which:

[0010] FIG. **1** is a top view of a battery case of a battery system according to some embodiments;

[0011] FIG. **2** is a cross-section view of the battery system of FIG. **1** taken along line **2-2** according to some embodiments;

[0012] FIG. **3**A is a zoomed-in view of area **3**A of FIG. **2** according to some embodiments;

[0013] FIG. **3**B is a perspective view illustrating portions of a thermal protection system for a battery system according to some embodiments;

[0014] FIG. **3**C is an end view illustrating portions of a thermal protection system for a battery system according to some embodiments;

[0015] FIG. **4** is a cross-section view of the battery system of FIG. **1** taken along line **4-4** according to some embodiments;

[0016] FIG. **5** is a cross-section view of the battery system of FIG. **1** taken along line **5-5** according to some embodiments;

[0017] FIG. **6** is a perspective view illustrating portions of a thermal protection system for a battery system according to some embodiments;

[0018] FIG. **7** is a zoomed-in view of area **7** of FIG. **6** according to some embodiments;

[0019] FIG. **8** is a top view of a battery case of a battery system according to some embodiments;

[0020] FIG. **9** is a cross-section view of the battery system of FIG. **8** taken along line **9-9** according to some embodiments;

[0021] FIG. **10** is a zoomed-in view of area **10** of FIG. **9** according to some embodiments;

[0022] FIG. **11** is a cross-section view of the battery system of FIG. **8** taken along line **11-11** according to some embodiments;

[0023] FIG. **12** is a top view of a battery case of a battery system according to some embodiments;

[0024] FIG. **13** is a cross-section view of the battery system of FIG. **12** taken along line **13-13** according to some embodiments;

[0025] FIG. **14** is a zoomed-in view of area **14** of FIG. **13** according to some embodiments;

[0026] FIG. **15** is a cross-section view of the battery system of FIG. **12** taken along line **15-15** according to some embodiments;

[0027] FIG. **16** is a top view of a battery case of a battery system according to some embodiments;

[0028] FIG. **17** is a cross-section view of the battery system of FIG. **16** taken along line **17-17** according to some embodiments;

[0029] FIG. **18** is a cross-section view of the battery system of FIG. **16** taken along line **18-18** according to some embodiments; and

[0030] FIGS. **19**A and **19**B are flow chart diagrams illustrating a method of assembling a thermal runaway protection system for a battery according to some embodiments.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0031] Referring now to the drawings, wherein like reference numbers are used herein to designate like or similar elements throughout the various views, illustrative embodiments are shown and described. The figures are not necessarily drawn to scale, and in some instances the drawings can be exaggerated or simplified in places for illustrative purposes only. One of ordinary skill in the art can appreciate many possible applications and variations for other embodiments based on the following illustrative embodiments provided in the present disclosure.

[0032] Some embodiments of the present disclosure may be intended to solve or address thermal management issues that might occur in the life cycle of a propulsion battery on an aircraft or in a vehicle. Some embodiments of the present disclosure may prevent or reduce module-to-module thermal runaway propagation by utilizing the abrupt pressure increase in the battery module enclosure driven by exothermic chemical reactions of pouch cells. The fluid energy generated under pressure inside a battery module enclosure may be partially converted into other forms of energy to detect and create a separation air gap that may act as a thermal barrier between an overheating battery module enclosure and its adjacent battery module enclosure(s).

[0033] Battery systems may incorporate a variety of devices (passive, active, or both) that detect, contain, and protect against thermal runaway issues. A thermal runaway protection system according to some embodiments of the present disclosure may be incorporated alone or in conjunction with other battery protection devices or systems, for example.

[0034] Battery systems having multiple battery cells or packs combined to form an effectively larger battery or an array of multiple batteries are used in many different applications, such as those using lithium-ion batteries, for example. Although an example embodiment of a battery system used in an aircraft is discussed herein, some embodiments may be adapted or configured to be used for other applications and uses, as may be apparent to one of ordinary skill in the art to which the present disclosure pertains, including systems and devices using multiple battery cells or packs, such as electric vehicles, automobiles, spacecraft, trains, boats, construction equipment, commercial equipment, industrial equipment, and consumer products, for example. Some embodiments have potential for usage on electric vehicles (EV) including battery electric aircraft (BEA) and hybrid electric aircraft (HEA) in the general aviation segment, regional aircraft, and large aircraft categories, for example.

[0035] Some embodiments use a protection structure to separate an overheated or overheating battery module from adjacent battery modules to provide some additional thermal insulation between the overheated or overheating battery module and the other battery modules. Some embodiments make use of a piston to move and separate the overheated or overheating battery module in response to the piston being acted upon by pressure that rises in a battery module enclosure due to the overheating. When such pressure acting on the piston is strong enough, it may overcome a force acting on the piston by a spring.

[0036] FIG. **1** is a top view of a battery case **30** of a battery system **32** according to some embodiments. FIG. **2** is a cross-section view of the battery system **32** of FIG. **1** taken along line **2-2**. FIG. **3**A is a zoomed-in view of area **3**A of FIG. **2**. FIG. **3**B is a perspective view of a protective structure **34** and piston **40** according to some embodiments. FIG. **3**C is an end view of a protective structure **34** and piston **40** according to some embodiments. FIG. **4** is a cross-section view of the battery system **32** of FIG. **1** taken along line **4-4**. FIG. **5** is a cross-section view of the battery system **32** of FIG. **1** taken along line **5-5**. FIG. **6** is a perspective view illustrating portions of a thermal runaway protection system **38** for a battery system **32** according to some embodiments. FIG. **7** is a zoomed-in view of area **7** of FIG. **6**.

[0037] In the example battery system **32** illustrated in FIGS. **1-7** in various views and perspectives, according to some embodiments, the battery system **32** includes three battery module enclosures **46** in a battery case. Each battery module enclosure **46** may include a set or pack of pouch cells, for example. The battery module enclosure **46** may be partially, substantially, or fully hermetically sealed to bound an enclosure interior space **48** with pouch cells therein, to protect the battery cells therein from external elements and contaminants, and to prevent or hinder gases or chemical elements of the battery cells therein from exiting the battery module enclosure **46**.

[0038] Although the example battery system **32** illustrated in FIGS. **1-7** shows battery modules in a battery case, some embodiments of the present disclosure can be implemented for a battery pack, a battery enclosure, or any compartment that is considered as a space inside or outside an aircraft where a battery fire may occur or may need to be contained, hindered, or prevented.

[0039] A battery module enclosure **46** may be made from a mica laminated ceramic paper core sandwiched between layers of carbon fiber, for example. A battery module enclosure **46** may be made from material(s) having good thermal conduction properties and ability to form a sealed and structurally sound enclosure for the battery pouches or cells therein, for example. A battery module enclosure **46** may be made from or include any suitable material, including but not necessarily limited to metal, steel, stainless steel, aluminum, titanium, bronze, brass, nickel, carbon fiber, fiberglass, titanium-carbon-fiber composite, structural composite material, carbon fiber reinforced polymers (CFRP), forged carbon composite, carbon nanotube composites (CNT), fiberglass reinforced polymers (FRP), glass fiber reinforced polymer (GFRP), aramid fiber composite, natural fiber composites, basalt fiber composite, metal matrix composite, woven composite, ceramic matrix composite (CMC), thermoplastic composites, various alloys thereof, various forgings thereof, or combinations thereof, for example. A battery module enclosure **46** may be made from multiple pieces layered, bonded, attached, threaded, welded, or fastened together, and a battery module enclosure **46** may be made from multiple materials combined, layered, bonded, attached, fastened, or any combination thereof, for example.

[0040] As illustrated in FIGS. **2**, **4**, and **5**, for example, the battery case **30** may include air spaces **44** inside the battery case **30** on the outside of the battery module enclosures **46**, which may be used for insulating the battery module enclosures **46** from external temperatures outside the battery case **30**. For example, during use, operation, loading, unloading, and parking of an aircraft, the aircraft and components thereof may be exposed to a wide variety of weather conditions, such as very hot, very cold, humid, dry, rainy, snowy, icy, or combinations thereof. Also, the air spaces inside the battery case **30** may be ventilated, coupled to a heating system, coupled to a cooling system, or combinations thereof, for adjusting or controlling temperature conditions experienced by the battery system **32** in the battery case **30**, for example.

[0041] The configuration and positions of the battery module enclosures **46** shown in FIGS. **1-7** may be a default or normal operating configuration and positions for the battery module enclosures **46** during typical use, for example. In some embodiments, the battery module enclosures **46** may be separated by divider walls **50** located between the battery module enclosures **46**. Such divider walls **50** may be structured, configured, and made from material(s) that function as a heat insulator by incorporating insulating material, for example. Such divider walls **50** may be made from insulating material, semiconducting material, conducting material, or combinations thereof, for example. A divider wall **50** may be made from a mica laminated ceramic paper core sandwiched between layers of carbon fiber, for example. A divider wall **50** may be made from or include any suitable material, including but not necessarily limited to metal, steel, stainless steel, aluminum, titanium, bronze, brass, nickel, carbon fiber, fiberglass, titanium-carbon-fiber composite, structural composite material, carbon fiber reinforced polymers (CFRP), forged carbon composite, carbon nanotube composites (CNT), fiberglass reinforced polymers (FRP), glass fiber reinforced polymer (GFRP), aramid fiber composite, natural fiber composites, basalt fiber composite, metal matrix composite, woven composite, ceramic matrix composite (CMC), thermoplastic composites, various alloys thereof, various forgings thereof, or combinations thereof, for example. A divider wall **50** may be made from multiple pieces layered, bonded, attached, threaded, welded, or fastened together, and a divider wall **50** may be made from multiple materials combined, layered, bonded, attached, fastened, or any combination thereof, for example.

[0042] In some embodiments, as illustrated in FIGS. **1-7**, the default or normal operating configuration may be having the divider walls **50** sandwiched between adjacent battery module enclosures **46**, with the battery module enclosures **46** proximate to, immediately proximate to, or even in contact with the divider wall **50** therebetween, for example. For various design and usage reasons, it may be desirable to retain the battery module enclosures **46** sandwiched together during normal operation of an aircraft or vehicle, such as for weight distribution or weight ballasting, for example.

[0043] In the event of a thermal runaway of one of the battery cells in a given battery module enclosure **46**, or in response to detecting conditions indicating a potential thermal runaway or start of a thermal runaway condition, some embodiments may provide a way to prevent, hinder, protect against, or combinations thereof, a thermal runaway event in one or more of the battery module enclosures **46**. In some embodiments, a thermal runaway event or conditions may be detected and contained. If one battery cell is detected as having a problem or starting thermal runaway conditions or already having thermal runaway conditions, it may be desired or required to electrically disengage and physically separate that battery module enclosure from other battery modules to protect the other battery modules that are still in operation and still functioning properly.

[0044] In some embodiments, one or more protection structures **34** of a thermal runaway protection system **38** may be incorporated into each of the battery module enclosures **46**, fluidly communicating with the enclosure interior space **48** within each battery module enclosure **46**. As illustrated in FIGS. **2** and **3**A, the protection structure **34** may include a first chamber inner surface **51** proximate a first structure end **61**. The first chamber inner surface **51** may partially bound a first chamber interior space **71** inside the protection structure **34**. The first chamber interior space **71** may have a first chamber width **81**. As illustrated in FIG. **3**A, the protection structure **34** may include a second chamber inner surface **52** at a second structure end **62**. The second chamber inner surface **52** may partially bound a second chamber interior space **72** having a second chamber width **82**. In some embodiments, the first chamber width **81** may be greater than the second chamber width **82**.

[0045] In some embodiments, the second structure end **62** of the protective structure **34** may be attached to the battery module enclosure **46**, as separate pieces joined or fastened together, as illustrated in FIG. **3**A for example. The protective structure **34** may be attached to the battery module enclosure **46** by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example. In some embodiments, part, parts, or all of the protection structure **34** and the battery module enclosure **46** may be a single integral part, such as being molded together, as opposed to multiple parts.

[0046] As illustrated in FIGS. **2**, **3**A, and **3**B, the protection structure **34** may include a pressure hole **84** at the first structure end **61**, formed in and through the first structure end **61**, and which may fluidly communicate with the enclosure interior space **48** inside the battery module enclosure **46**. The pressure hole **84** may have a third width **83**. The first chamber width **81** may be greater than the third width **83** of the pressure hole **84**. The protection structure **34** may include a set of vent holes **88** at the second structure end **62**, formed in and through the second structure end **62**, which may fluidly communicate with the first chamber interior space **71**, and which may fluidly communicate with a space between the divider wall **50** and the battery module enclosure **46** for which the protective structure **34** is attached or included. The set of vent holes **88** may extend through the second structure end **62** and be disposed proximate to, and outside of the second chamber inner surface **52**. In some embodiments, the vent holes **88** may be disposed radially around the second chamber inner surface **52**. In some embodiments, the set of vent holes **88** may be a series of evenly spaced and distributed circular holes molded or drilled in the second structure end **62**. In other embodiments, the number, shape, size, through angle, and configuration of the vent hole(s) **88** may vary from the examples shown in FIGS. **2**, **3**A, and **3**C.

[0047] As illustrated in FIGS. **2** and **3**A, the protection structure **34** may include a set of lock pawl pockets **94** in the first chamber inner surface **51**, which may be molded or machined therein as blind holes, recessed areas or regions, or recessed cavities, for example. In some embodiments, the number, shape, size, wall angles, and configuration of the lock pawl pocket(s) **94** may vary from the examples shown in FIGS. **2** and **3**A.

[0048] As illustrated in FIGS. **2**, **3**A, **3**B, and **3**C, a piston **40** may be located in the protection structure **34**. The piston **40** may include a first piston outer surface **91** at a first piston end **41**. The first piston end **41** may have a first piston end width **101**. The piston **40** may include a second piston outer surface **92** at a second piston end **42**. The second piston end **42** may have a second piston end width **102**. The first piston end width **101** may be greater than the second piston end width **102**. The first piston end **41** may fit in and slidably couple with the first chamber inner surface **51**, such that the first piston end **41** is in the first chamber interior space **71**. The second piston end **42** may fit in and slidably couple with the second chamber inner surface **52**, such that the second piston outer surface **92** may be proximate to, or in contact with, the second chamber inner surface **52**, and such that the second piston end **42** is in the second chamber inner space **72**, as illustrated in FIG. **3**A for example.

[0049] The second piston end **42** may extend outside the protection structure **34** and may be attached to the divider wall **50**, by using a bolt, rivet, screw, or other fastener or other fastening technique such as bonding, welding, adhesive, or the like, or any combination thereof, for example. Hence, a piston **40** and a divider wall **50** may be configured to move together relative to a protection structure **34** and a battery module enclosure **46**.

[0050] At least part of the first piston end **41** may fluidly communicate with the enclosure interior space **48** via the pressure hole **84**. The first piston outer surface **91** may contact the first chamber inner surface **51** such that pressure within the enclosure interior space **48** may be hindered or prevented from exiting the enclosure interior space **48** past the first piston end **41** and such that pressure within the enclosure interior space **48** may press against the first piston end **41** via the pressure hole **84**.

[0051] As illustrated in FIGS. **3**B and **3**C, an exterior of a protection structure **34**, interior chamber(s) of a protection structure **34**, a piston **40**, or any combination thereof may have a cylindrical shape with a circular cross-section. In some embodiments, an exterior of a protection structure **34**, interior chamber(s) of a protection structure **34**, a piston **40**, or any combination thereof may have any suitable shape, such as having a rectangular, square, oval, or hexagon shaped cross-section, for example.

[0052] A piston **40** may be made from titanium for some aircraft applications, for example. A piston **40** may be made from any suitable material, including but not necessarily limited to metal, steel, stainless steel, aluminum, titanium, bronze, brass, nickel, carbon fiber, fiberglass, titanium-carbon-fiber composite, structural composite material, carbon fiber reinforced polymers (CFRP), forged carbon composite, carbon nanotube composites (CNT), fiberglass reinforced polymers (FRP), glass fiber reinforced polymer (GFRP), aramid fiber composite, natural fiber composites, basalt fiber composite, metal matrix composite, woven composite, ceramic matrix composite (CMC), thermoplastic composites, various alloys thereof, various forgings thereof, or combinations thereof, for example. A piston **40** may be a single integral piece of a same material, or a piston **40** may be made from multiple pieces attached, threaded, welded, or fastened together, and a piston **40** may be made from multiple materials combined, layered, bonded, attached, fastened, or any combination thereof, for example.

[0053] A protection structure **34** may be made from any suitable material, including but not necessarily limited to metal, steel, stainless steel, aluminum, titanium, bronze, brass, nickel, carbon fiber, fiberglass, titanium-carbon-fiber composite, structural composite material, carbon fiber reinforced polymers (CFRP), forged carbon composite, carbon nanotube composites (CNT), fiberglass reinforced polymers (FRP), glass fiber reinforced polymer (GFRP), aramid fiber composite, natural fiber composites, basalt fiber composite, metal matrix composite, woven composite, ceramic matrix composite (CMC), thermoplastic composites, various alloys thereof, various forgings thereof, or combinations thereof, for example. A protection structure **34** may be a single integral piece of a same material, or a protection structure **34** may be made from multiple pieces attached, threaded, welded, or fastened together, and a protection structure **34** may be made from multiple materials combined, layered, bonded, attached, fastened, or any combination thereof, for example.

[0054] As illustrated in FIGS. **2**, **3**A, and **3**C, a spring **104** may be located in a first chamber interior space **71** surrounding a piston **40**. The spring **104** may be compressed between or configured to be biased between a first piston end **41** and a second structure end **62**. In some embodiments, the spring **104** may overlap with the vent holes **88** without fully blocking the vent holes. In some embodiments, a spring **104** and vent hole(s) **88** may be sized, configured, and positioned such that the spring **104** does not overlap with the vent hole(s) **88** at all or not as much as illustrated in FIG. **3**C.

[0055] A spring **104** may be sized and configured to provide a specified force against the first piston end **41** to counteract a specified level or range of pressure in the enclosure interior space **48** to counter a pressure force exerted on the first piston end **41** via the pressure hole **84**. The pressure may arise in the enclosure interior space **48** because such space is sealed (constant volume) and the temperature increases, such as during a thermal runaway event. Thus, the increased pressure in the enclosure interior space **48** may correlate with the increased temperature of the battery cells in the battery module enclosure. Hence, the sizing, material(s), and configuration of the spring **104** in combination with other factors, such as but not necessarily limited to the size of the pressure hole **84**, the friction of the first piston outer surface **91** against the first chamber inner surface **51**, and the friction of the second piston outer surface **92** against the second chamber inner surface **52**, may be tuned to hinder or prevent movement of the piston **40** due to pressure below a specified or selected threshold within the enclosure interior space **48** acting on the first piston end **41** via the pressure hole **84**. And likewise, the sizing, material(s), and configuration of the spring **104**, combined with other factors, may be tuned to allow movement of the piston **40** due to pressure above a specified or selected threshold within the enclosure interior space **48** acting on the first piston end **41** via the pressure hole **84**. A spring **104** may be made from stainless steel, for example.

[0056] As illustrated in FIGS. **2** and **3**A, a set of lock pawls **106** may be located in the set of lock pawl pockets **94**. Each of the lock pawls **106** may have a sloped shape configured to allow the first piston end **41** to press the lock pawls **106** into the lock pawl pockets **94** while the piston **40** moves in a first direction **111**, as illustrated in FIGS. **10** and **14**) while compressing the spring **104**. Also, each of the lock pawls **106** may have a ratchet tooth structure configured to hinder or prevent the piston **40** from moving in a second direction **112** after the first piston end **41** passes the ratchet tooth structure while the piston **40** moves in the first direction **111**. The first direction **111**, compressing the spring **104**, is opposite the second direction **112**, where moving the piston **40** in the second direction **112** allows the spring **104** to expand, as illustrated in FIG. **3**A. In some embodiments, when a piston **40** is moved by pressure acting upon it and the force of the spring **104** is overcome to an extent that the piston **40** moves beyond a set of lock pawls **106**, the lock pawls may restrict the piston **40** from in the second direction. Accordingly, the spring **104** may press the piston **40** against the lock pawls **106** to lock in place the position of the piston, and thereby retain a separation between adjacent battery module enclosures. Each of the lock pawls **106** may be made of a material and may have a structure configured to elastically return to an original position partially extending outside the respective lock pawl pocket **94** after being pressed into the lock pawl pocket **94** by the first piston end **41**.

[0057] A lock pawl **106** may be made from any suitable material, including but not necessarily limited to metal, steel, stainless steel, aluminum, titanium, bronze, brass, nickel, carbon fiber, fiberglass, titanium-carbon-fiber composite, structural composite material, carbon fiber reinforced polymers (CFRP), forged carbon composite, carbon nanotube composites (CNT), fiberglass reinforced polymers (FRP), glass fiber reinforced polymer (GFRP), aramid fiber composite, natural fiber composites, basalt fiber composite, metal matrix composite, woven composite, ceramic matrix composite (CMC), thermoplastic composites, various alloys thereof, various forgings thereof, or combinations thereof, for example. A lock pawl **106** may be a single integral piece of a same material, or a lock pawl **106** may be made from multiple pieces attached, welded, or fastened together, and a lock pawl **106** may be made from multiple materials combined, layered, bonded, attached, fastened, or any combination thereof, for example. A lock pawl **106** may be held in place in its respective lock pawl pocket **94** by fastener, adhesive, bonding, welding, wedge fit, slip fit, loose fit, clip structure, or any combination thereof, for example.

[0058] As illustrated in FIGS. **4-7**, for example, in some embodiments, tracks **121**, **122** and track wheels **151**, **152** may be included to guide, control, and reduce friction of movement of one or more battery module enclosures **46**. In some embodiments, a first track **121** may extend across a first battery case width **131** proximate to a first battery case end **141**, for example. The first track **121** may be attached to the battery case **30** by using bolts, rivets, screws, brackets, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example. In some embodiments, a second track **122** may extend across a second battery case width **132** proximate to a second battery case end **142**, for example. The second track **122** may be attached to the battery case **30** by using bolts, rivets, screws, brackets, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example. A track **121**, **122** may be made from any suitable material, including but not necessarily limited to metal, steel, stainless steel, aluminum, titanium, bronze, brass, nickel, carbon fiber, fiberglass, titanium-carbon-fiber composite, structural composite material, carbon fiber reinforced polymers (CFRP), forged carbon composite, carbon nanotube composites (CNT), fiberglass reinforced polymers (FRP), glass fiber reinforced polymer (GFRP), aramid fiber composite, natural fiber composites, basalt fiber composite, metal matrix composite, woven composite, ceramic matrix composite (CMC), thermoplastic composites, various alloys thereof, various forgings thereof, or combinations thereof, for example.

[0059] In some embodiments, as illustrated for example in FIGS. **4**, **6**, **7**, **11**, **15**, and **18**, a first set of first end track wheels **151** may be attached to a first end **161** of a first battery module enclosure **46***a* along with or via a first track wheel bracket **171**, for example. The first set of first end track wheels **151** may be engaged and slidably coupled with the first track **121**. The first set of first end track wheels **151**, the first track wheel bracket **171**, or both may be attached to the first battery module enclosure **46***a* by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example. In some embodiments, track wheels and tracks may be used to guide and reduce friction of movement of the overheated battery module enclosure. For example, the track wheels and tracks may reduce or prevent stiction, binging, or interference that may otherwise prevent or hinder the movement of the overheated battery enclosure.

[0060] In some embodiments, as illustrated for example in FIG. **5**, a second set of first end track wheels **151** may be attached to a first end **161** of a second battery module enclosure **46***b* along with or via a second track wheel bracket **172**, for example. The second set of first end track wheels **151** may be engaged and slidably coupled with the first track **121**. The second set of first end track wheels **151**, the second track wheel bracket **172**, or both may be attached to the second battery module enclosure **46***b* by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example.

[0061] In some embodiments, a first dividing wall bracket **181** may be attached to a first end of the dividing wall **50**, for example. The first dividing wall bracket **181** may be engaged and slidably coupled with the first track **121**. The first dividing wall bracket **181** may be attached to the dividing wall **50** by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example. In some embodiments, the first dividing wall bracket **181** can be used to guide and allow the divider wall **50** to move more freely with less friction.

[0062] In some embodiments, a first set of second end track wheels **152** may be attached to a second end **162** of the first battery module enclosure **46***a* along with or via a third track wheel bracket **173**, for example. The first set of second end track wheels **152** may be engaged and slidably coupled with the second track **122**. The first set of second end track wheels **152**, the third track wheel bracket **173**, or both may be attached to the first battery module enclosure **46***a* by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding,

adhesives, or the like, or any combination thereof, for example.
[0063] In some embodiments, a second set of second end track wheels **152** may be attached to a second end **162** of the second battery module enclosure **46***b* along with or via a fourth track wheel bracket **174**, for example. The second set of second end track wheels **152** may be engaged and slidably coupled with the second track **122**. The second set of second end track wheels **152**, the fourth track wheel bracket **174**, or both may be attached to the second battery module enclosure by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example.
[0064] In some embodiments, a second dividing wall bracket **182** may be attached to a second end of the dividing wall **50**, for example. The second dividing wall bracket **182** may be engaged and slidably coupled with the second track **122**. The second dividing wall bracket **182** may be attached to the dividing wall **50** by using bolts, rivets, screws, or other fasteners or other fastening techniques such as bonding, welding, adhesives, or the like, or any combination thereof, for example. In some embodiments, the second dividing wall bracket **182** can be used to guide and allow the divider wall **50** to move more freely with less friction.
[0065] Although a track and wheel configuration is used in some example embodiments, other guide structures and configurations may be implemented in some embodiments, such as a tongue-in-groove configuration, a slider bracket, a slider mechanism, rail-in-groove, runners, ball-bearing slides, roller-bearing slides, friction slides, low-friction-layer-coated friction slides, lubricated slides, dovetail slides, rack and pinion gears, or any combination thereof, for example.
[0066] FIGS. **1-7** illustrate the battery system **32** in a default or normal operation configuration, such as before a thermal runaway event has occurred for example, according to some embodiments. FIGS. **8-17** illustrate a battery system **32** after a thermal runaway detection and containment, or protection, system **38** has been activated by increased pressure in an enclosure interior space **48** of a given battery module enclosure **46** for the battery system **32**, according to some embodiments, which illustrate some example scenarios.
[0067] FIG. **8** is a top view of a battery case **30** of a battery system **32** according to some embodiments. FIG. **9** is a cross-section view of the battery system **32** of FIG. **8** taken along line **9-9**. FIG. **10** is a zoomed-in view of area **10** of FIG. **9**. FIG. **11** is a cross-section view of the battery system **32** of FIG. **8** taken along line **11-11**. FIGS. **8-11** illustrate an example scenario in which increased pressure in an enclosure interior space **48** of a battery module enclosure **46***a* shown on a left side in the figures has triggered a thermal runaway detection and containment system **38** according to some embodiments.
[0068] FIG. **12** is a top view of a battery case **30** of a battery system **32** according to some embodiments. FIG. **13** is a cross-section view of the battery system **32** of FIG. **12** taken along line **13-13**. FIG. **14** is a zoomed-in view of area **14** of FIG. **13**. FIG. **15** is a cross-section view of the battery system **32** of FIG. **12** taken along line **15-15**. FIGS. **12-15** illustrate an example scenario in which increased pressure in an enclosure interior space **48** of a battery module enclosure **46***c* shown on a right side in the figures has triggered a thermal runaway detection and containment system **38** according to some embodiments.
[0069] FIG. **16** is a top view of a battery case **30** of a battery system **32** according to some embodiments. FIG. **17** is a cross-section view of the battery system **32** of FIG. **16** taken along line **17-17**. FIG. **18** is a cross-section view of the battery system **32** of FIG. **16** taken along line **18-18**. FIGS. **16-18** illustrate an example scenario in which increased pressure in an enclosure interior space **48** of a battery module enclosure **46***b* located in a center position has triggered a thermal runaway detection and containment system **38** according to some embodiments.
[0070] When temperature and pressure increase inside a battery module enclosure **46** and the pressure in the enclosure interior space **48** exceeds a selected or specified threshold, the pressure in the enclosure interior space **48** acting upon the first piston end **41** via the pressure hole **84** may overcome the force of the spring **104**, and other friction forces of the structures and components, to

move the piston **40** relative to the protection structure **34**. An increased pressure in an enclosure interior space **48** of a battery module enclosure **46**, which may be caused by a thermal runaway scenario for example, has pressed on a first piston end **41** of the piston **40** via the pressure hole **84**, thereby moving the piston **40** in a first direction **111** by overcoming the forces provided by the spring **104**, various friction points for components in contact with each other, and inertia of the weight of the components being moved. As the piston **40** moves in the first direction **111**, the piston **40** may encounter the lock pawls **106**. Due to the sloped shape of the lock pawls **106**, as the piston **40** engages with lock pawls **106** while moving in the first direction **111**, the first piston end **41** may press the lock pawls **106** into the lock pawl pockets **94**, which may enable the piston **40** to continue moving in the first direction **111**. After the first piston end **41** passes the lock pawls **106** while continuing to compress the spring **104** and moving in the first direction **111**, the lock pawls **106** may elastically return to a position in which all of or at least one of the lock pawls **106** again extend out from the first chamber inner surface **51** and the ratchet tooth structure of all or at least one of the lock pawls **106** may hinder or prevent the movement of the piston **40** in the second direction.

[0071] The ratchet tooth structure of the lock pawls **106** may block a path of the first piston end **41** from moving in the second direction after the first piston end **41** passes the ratchet tooth structure while the piston **40** moves in the first direction **111** because the lock pawl **106** may spring back or elastically return to a position such that the lock pawl **106** extends out of or at least partially extends outside of the respective lock pawl pocket **94**. After the first piston end **41** passes the lock pawls **106**, the lock pawls **106** may block, hold, or lock the piston **40** at a locked position, as illustrated in FIGS. **10** and **14** for example.

[0072] A separation air gap **90** may form between a divider wall **50** and a battery module enclosure **46** when the piston **40** moves in the first direction **111** compressing the spring **104** and when the ratchet tooth structures of the lock pawls **106** hinder the piston **40** from moving in the second direction. During such movement of the piston **40** in the first direction **111**, fluid, air, or gas in the first chamber interior space **71** may be vented out of the first chamber interior space **71** via the vent holes **88** and into the separation air gap **90** formed because the vent holes **88** may fluidly communicate with the first chamber interior space **71** and the separation air gap **90**. The first chamber interior space **71** may fluidly communicate with the separation air gap **90** via the vent holes **88**.

[0073] The separation air gap **90**, may provide an insulation layer between the battery module enclosure **46** and the divider wall **50**, which may contain or help contain increased temperatures in the battery module enclosure experiencing a thermal runaway condition, and which may thermally insulate an overheated battery module enclosure from other battery module enclosures in the battery case **30**. A battery module enclosure experiencing a thermal runaway condition may be electrically disconnected from system(s) or device(s) making used of the batteries in that battery system, by some other safety system for example, but at the same time or in addition or in alternative, the increased temperature generated in that overheated battery module enclosure may be thermally isolated or insulated from other battery module enclosures in the same battery case **30** that may be still operating and may be still used.

[0074] In some embodiments, the track wheel brackets **171**, **172**, **173**, **174** may be omitted, as an optional feature or component. In some embodiments, the dividing wall brackets **181**, **182** may be omitted, as an optional feature or component. For example, in some embodiments the divider wall **50** may be permitted or expected to move under certain circumstances. And in some embodiments, the divider wall **50** may be fixed or attached to the battery case **30** such that the divider wall does not move or is not intended to move during normal operation or even when a battery module enclosure **46** moves, for example.

[0075] The friction of the dividing wall brackets **181**, **182** sliding on the tracks **121**, **122**, the friction of the battery module enclosure **46** sliding along inner surfaces of the battery case **30**, if they are touching, the friction of the track wheels **151**, **152** rotating along the tracks **121**, **122**, or any combination thereof, along with other factors, such as the size of the pressure hole **84**, the friction of the first piston outer surface **91** against the first chamber inner surface **51**, the friction of the second piston outer surface **92** against the second chamber inner surface **52**, if they are touching, or any combination thereof, may be considered and used to select and tune a size, material(s), and configuration of the spring **104** for countering and allowing movement of the piston **40** for specified or selected pressure thresholds or ranges. In some embodiments, multiple springs or a spring with a variable spring rate, or combinations thereof, may be used in place of or in alternative of a single spring **104**.

[0076] A size of the piston **40** relative to a size of a battery module enclosure **46** may depend on the force required to separate the battery module enclosure **46** from the divider wall **50** because such force may be proportional to the piston's cross-sectional area and triggering pressure. Due to weight constrains in aircraft, for example, it may be recommended to design the piston **40** as small as feasible relative to the battery module enclosure **46**, for example.

[0077] In some embodiments, the dividing wall **50** may be moveable and may be intended to move when a thermal runaway protection system **38** is triggered or activated. In some embodiments, one or more of the dividing walls **50** of a battery system **32** may be attached to the battery case **30** or may be fixed relative to the battery case **30**, such that when a thermal runaway protection system **38** is triggered or activated, only an overheated battery module enclosure **46** may move to form a separation air gap **90** between the overheated battery module enclosure and the fixed divider wall.

[0078] FIGS. **19**A and **19**B are flow chart diagrams illustrating a method of assembling a thermal runaway protection system **38** for a battery system **32** according to some embodiments. At operation **1905**, a battery module enclosure **46** including an enclosure interior space **48** may be provided. In some embodiments, the battery module enclosure **46** may be provided by sourcing, procurement, partial manufacturing of a sourced or procured part, full manufacturing, or some combination thereof, for example.

[0079] At operation **1910**, a divider wall **50** may be provided. In some embodiments, the divider wall **50** may be provided by sourcing, procurement, partial manufacturing of a sourced or procured part, full manufacturing, or some combination thereof, for example.

[0080] At operation **1915**, a protection structure **34** including a first chamber inner surface **51** proximate to a first structure end **61** may be provided. The first chamber inner surface **51** may partially bound a first chamber interior space **71**. The protection structure **34** may further include a pressure hole **84** at the first structure end **61**. The protection structure **34** may further include a set of vent holes **88** through a second structure end **62**. The vent holes **88** may fluidly communicate with the first chamber interior space **71**. The protection structure **34** may further include a set of lock pawl pockets **94** in the first chamber inner surface **51**. In some embodiments, the protection structure **34** may be provided by sourcing, procurement, partial manufacturing of a sourced or procured part, full manufacturing, or some combination thereof, for example.

[0081] At operation **1920**, a piston **40** including a first piston outer surface **91** at a first piston end **41** may be provided. The piston **40** may further include a second piston outer surface **92** at a second piston end **42**. The first piston end **41** may have a first piston end width **101**. The second piston end **42** may have a second piston end width **102**. The first piston end width **101** may be greater than the second piston end width **102**. In some embodiments, the piston **40** may be provided by sourcing, procurement, partial manufacturing of a sourced or procured part, full manufacturing, or some combination thereof, for example.

[0082] At operation **1925**, a set of lock pawls **106** may be placed in the set of lock pawl pockets **94**. In some embodiments, the set of lock pawls **106** may be provided by sourcing, procurement, partial manufacturing of a sourced or procured part, full manufacturing, or some combination thereof, for example.

[0083] At operation **1930**, a spring **104** may be placed in the first chamber interior space **71** inside the protection structure **34**. In some embodiments, the spring **104** may be provided by sourcing, procurement, partial manufacturing of a sourced or procured part, full manufacturing, or some combination thereof, for example.

[0084] At operation **1935**, the piston **40** may be placed into the protection structure **34** such that the spring **104** is surrounding the piston **40**, such that the first piston end **41** fits in and slidably couples with the first chamber inner surface **51**, such that the first piston outer surface **91** contacts the first chamber inner surface **51**, such that the first piston end **41** is in the first chamber interior space **71**, such that the spring **104** is configured to bias between the first piston end **41** and the second structure end **62**, and such that at least part of the first piston end **41** fluidly communicates with the enclosure interior space **48** via the pressure hole **84**. In some embodiments, the protection structure **34** may be two pieces that fit together to allow for inserting the lock pawls **106**, spring **104**, and piston **40** in a first half of the protection structure and then placing, fastening, threading, bonding, or attaching a second half of the protection structure on or to the first half of the protection structure, for example. One of ordinary skill in the art may realize many possible ways and configurations to construct and assemble the piston **40** and protection structure **34** together while placing the piston into the protection structure.

[0085] At operation **1940**, the second structure end **62** may be attached to the battery module enclosure **46**, such that the pressure hole **84** fluidly communicates with the enclosure interior space **48** of the battery module enclosure **46**.

[0086] At operation **1945**, the second piston end **42** may be attached to the divider wall **50**, such that the divider wall **50** is proximate to the battery module enclosure **46**.

[0087] At operation **1950**, a first dividing wall bracket **181** may be attached to a first end of the dividing wall **50**. At operation **1955**, a set of first end track wheels **151** may be attached to a first end **161** of the battery module enclosure **46**. At operation **1960**, the first dividing wall bracket **181** may be engaged and slidably coupled with a first track **121**. At operation **1965**, the set of first end track wheels **151** may be engaged and slidably coupled with the first track **121**.

[0088] At operation **1970**, a second dividing wall bracket **182** may be attached to a second end of the dividing wall **50**. At operation **1975**, a set of second end track wheels **152** may be attached to a second end **162** of the battery module enclosure **46**. At operation **1980**, the second dividing wall bracket **182** may be engaged and slidably coupled with a second track **122**. At operation **1985**, the set of second end track wheels **152** may be engaged and slidably coupled with the second track **122**.

[0089] At operation **1990**, the first track **121** may be attached to a first end **141** of a battery case **30**. At operation **1995**, the second track **122** may be attached to a second end **142** of the battery case **30**. According to some embodiments, the operations **1905** to **1995** may be sequentially performed, but are not necessarily sequentially performed. For example, the sequence of the operations may be changed, or at least two operations may be performed in parallel. According to some embodiments, some operations of **1905** to **1995** may be omitted or substituted for other operations.

[0090] Some embodiments of the present disclosure may generally relate to the general field of all-electric or hybrid electric aircraft and may relate specifically to propulsion batteries for aircraft or vehicles. Containment and prevention of the module-to-module thermal runaway propagation may be crucial for the safe operation of all-electric or hybrid electric aircraft during an abnormal condition. In the event of thermal runaway, tremendous amount of heat and pressure may be generated abruptly by the chemical compounds contained inside the battery cells. Depending on the cell chemistry and the ambient conditions, each cell may react differently to the heat and pressure dissipation. Under certain conditions, the pressure increment may cause fire or explosion leading to a catastrophic failure. Cell chemistries, such as lithium-ion, tend to have vigorous effects during thermal runaway. Propulsion battery cells may generate heat due to the irreversible energy dissipation under usual circumstances, such as flight conditions affected by weather conditions. The cell temperature inside the propulsion battery enclosure may be maintained to preserve the cell health. To facilitate the battery thermal management, various methods and technologies may be viable. A passive cooling method may eliminate or reduce the number of active elements, such as sensors, control systems, and active components to monitor and adjust cell temperature. In some embodiments, the battery module enclosure **46** may be a good thermal conductor, and there may be an interface between the cells, enclosure, and the outside ambient environment to transfer the thermal energy. Most of materials may be classified as either good conductors or poor conductor, such as an insulator. To facilitate dual functionality, in some embodiments, an alternative mechanism may be developed and implemented that allows the propulsion battery to be cooled passively during normal operations and reduce, or significantly reduce or prevent, module-to-module thermal propagation during abnormal conditions, such as thermal runaway.

[0091] In some embodiments, a thermal runaway containment mechanism (TRCM) **38** may include several mechanical components such as a cylinder or protection structure **34** with a cylindrical inner chamber, a piston **40** with connecting rod, a mechanical spring **104**, and a lock pawl **106**. The TRCM **38** may interconnect a battery module enclosure **46** and a divider wall **50** or an insulation material. During a normal operating condition, the piston **40** may remain in an extension position by the force of the spring **104**, for example. The piston **40** may be designed with a connecting rod, such as a second piston end **42**, that is attached directly to the insulation material or a divider wall **50**. The insulation material or divider wall **50** may mimic a semiconductor by enabling passive heat dissipation during normal operating conditions. In the event of thermal runaway, for example, the piston **40** may be forced to translate due to an abrupt pressure generated by exothermic reactions of the cells. This may lead to a movement of the piston **40**. Consequently, the piston may push the adjacent battery module enclosures apart such that the insulation material or divider wall **50** may be separated from the attached or adjacent battery module enclosure that is overheating. Additionally, another adjacent battery module, which is not overheating and is still operating normally, may be separated by physically moving the other battery module enclosure(s) away from the overheating battery enclosure that induced thermal runaway, for example. The cylinder or protection structure **34** may be designed to interlock the position after the retraction of the piston **40** using the mechanism of the lock pawls **106**. Due to the existence of the air gap **90** between the overheated battery module enclosure **46** and the insulation material or divider wall **50**, the thermal conductivity may be reduced significantly by further reducing the thermal conductivity between the overheated battery module enclosure and the divider wall.

[0092] The TRCM **38** may be attached to the battery module enclosure **46** such that a front area of the piston **40**, such as the first piston end **41**, may be exposed directly to an inner surrounding environment, such as the enclosure interior space **48**, of the battery module enclosure **46**, such as via the pressure hole **84** in the first structure end **61** of the protection structure **34**. This may allow the piston **40** and the spring **104** to absorb the inner pressure during thermal runaway. In addition, the cylinder or protection structure **34** of the TRCM **38** may be capable of releasing the back pressure within the first chamber interior space **71** of the protection structure **34**, which may be developed during piston movement, by utilizing the venting holes **88**.

[0093] During an abnormal condition, continuous power delivery may be needed, preferred, or desired to reduce the pilot workload in all-electric or hybrid electric aircraft. In a hybrid electric configuration, the propulsion battery may be intended to provide power during the failure of the primary energy source. This may allow the pilot to perform a safe emergency landing while the secondary energy storage system provides partial power to the engine. An all-electric aircraft may be designed to prevent failure of the complete energy storage system. To mitigate or prevent the catastrophic failure, the prevention of thermal propagation to the adjacent battery modules or compartments may be incorporated, according to some embodiments. In this case, the pilot may have enough time to perform an emergency landing, and ultimately, land the aircraft safely.

[0094] In some embodiments, a TRCM **38** may include components that provide multiple features to prevent catastrophic failure. Combined with a module separation mechanism, such as the protection structure **34**, piston **40**, spring **104**, and lock pawls **106**, for example, some embodiments of the present disclosure may include attaching, removably or placing adjacently, an insulation material, such as a divider wall **50**, to a battery module enclosure **46** to facilitate a passive cooling method. Passive cooling elements may enable the reduction of materials required to insulate a propulsion battery enclosure, which may save weight in a design, while also providing insulation during thermal runaway situations. In some embodiments, achieving such dual functionality may provide unique advantages where, in general, material tends to be either good conductor or poor conductor (insulator), such as graphite and Aerogel, respectively. Another advantage of the weight reduction of insulation material may be provided in some embodiments in which passive cooling elements allow elimination or reduction of excess component weight as compared to active cooling functions. Overall, a TRCM **38** in some embodiments may provide an advantage of being a simple yet effective solution for providing dual functional features that may be incorporated to any or most or many propulsion batteries, regardless of the design and use cases.

[0095] In some embodiments, a TRCM **38** may mainly include passive elements. An advantage of using passive elements may be that electrical redundancy is not required compared to an active element, which may lead to less weight, less cost, and less complexity in the design. Another advantage of some embodiments may be that a TRCM **38** may allow a propulsion battery to be easily certified, such as with the European Aviation Safety Agency (EASA), which is an agency that ensures air transport safety in civil aviation by establishing comprehensive requirements and testing procedures.

[0096] Together with the feature of separating the battery module enclosures **46** in some embodiments, a TRCM **38** may reduce the overall weight of the system compared to alternative active systems, as yet another advantage. Reducing weight may provide additional overhead to the energy that may be carried into the propulsion batteries, reducing excess battery weight may lead to increased aircraft range, endurance, and other electrical performance parameters, for example.

[0097] Alternatively, a separation mechanism for separating one battery module enclosure from another battery module enclosure may be designed using a pressure transducer and other mechanical mechanisms to separate the adjacent propulsion battery module enclosures. Active elements may require electrical components with redundancy features, which may require each component to be redundant for some certifications. Also, additional components to facilitate the separation of the battery module enclosures may include microcontrollers and data acquisition system to monitor and process sensor data. In such active system, such components may increase the complexity as well as the weight of the overall system. Additional weight penalty may have significant impact on the aircraft performance.

[0098] During normal operating conditions, heat generated by a battery often should be dissipated to improve battery life and prevent potential overheating. With the aid of a passive cooling method, in some embodiments the heat may be conducted from the battery module enclosure **46** to the outside environment. In some embodiments, a TRCM **38** works together with the insulation material of a divider wall **50** attached to the piston **40** to dissipate the heat, because the battery module enclosure incorporates a thermal interface between the cells therein and the battery module enclosure. In some embodiments, a TRCM **38** in the non-triggered state, as illustrated in FIGS. **2-7** for example, due to the absence of high pressure in the enclosure interior spaces of the battery module enclosures, the mechanical spring **104** pressing on the piston **40** may prevent the piston from moving into a retracted or locked position. The spring **104** may prevent unwanted piston translation when a sudden pressure change occurs during flight of an aircraft for reasons other than thermal runaway, for example. The piston **40** and the protection structure **34** may be made of materials that can withstand high temperatures, for example. The outer enclosure or battery case **30** may be made of materials that are able to conduct the heat from the divider wall **50**, which may include an insulation material, and which may be attached to the second piston end **42** of the TRCM **38**. During normal operating conditions, the divider wall **50** may be contacting, against, touching, or proximate to an adjacent battery module enclosure **46**, as illustrated in FIGS. **2**, **3**A,

and **4-7** for example.
[0099] When thermal runaway occurs, the excess pressure may cause the piston **40** to move into a retraction position, as illustrated in FIGS. **9**, **10**, **13**, **14**, and **17** for example. In some embodiments, at a rear side of the cylinder, or at a second structure end **62** of a protection structure **34**, several venting holes **88** may release the back pressure of in the cylinder, or in the first chamber interior space **71**, which may allow the piston **40** to move to the retracted or locked position. After the piston movement, the spring **104** may be compressed. Once the piston **40** reaches the end of the lock pawl(s) **106**, the piston **40** may be remained in the retracted position, with the lock pawl(s) **106** preventing the piston **40** to move back to the initial position. This may allow the divider wall **50** to maintain a separated position while creating additional airgap **90** between the battery module enclosures **46**. Because air has lower thermal conductive properties than many solid materials, this separation air gap **90** may prevent or reduce the module-to-module thermal propagation.
[0100] An embodiment thermal runaway protection system for a battery includes a battery module enclosure, a divider wall, a protection structure, and a piston. In the embodiment, the battery module enclosure bounds an enclosure interior space. In the embodiment, the divider wall is adjacent to the battery module enclosure. In the embodiment, the protection structure is attached to the battery module enclosure. In the embodiment, the protection structure includes a first chamber inner surface proximate to a first structure end, where the first chamber inner surface at least partially bounds a first chamber interior space inside the protection structure, where the protection structure has a pressure hole at the first structure end, where the pressure hole is in fluid communication with the enclosure interior space of the battery module enclosure, where the protection structure has a vent hole through a second structure end, and where the vent hole is in fluid communication with the first chamber interior space. In the embodiment, the piston is at least partially disposed in the protection structure, where the piston at least partially separates the enclosure interior space from the first chamber interior space, where the piston has a first piston end and a second piston end, where the piston fits in, and slidably engages with, the first chamber inner surface, where at least part of the first piston end is in fluid communication with the enclosure interior space via the pressure hole, and where the second piston end is proximate to the divider wall.
[0101] In some embodiments, the system further includes a spring disposed between the first piston end and the second structure end, where the spring surrounds the piston in the first chamber interior space. In some embodiments, the system further includes a set of lock pawls, where the protection structure further includes a set of lock pawl pockets in the first chamber inner surface, and where the set of lock pawls are at least partially in the set of lock pawl pockets, where the piston includes a first piston outer surface at the first piston end, where the piston further includes a second piston outer surface at the second piston end, where the first piston end has a first piston end width greater than a second piston end width of the second piston end, where the first piston end fits in, and slidably engages with, the first chamber inner surface, where the first piston outer surface contacts the first chamber inner surface and the first piston end is disposed in the first chamber interior space, where the second piston end is attached to the divider wall, and where the second structure end is attached to the battery module enclosure.
[0102] In some embodiments, the first chamber interior space has a first chamber width. In some embodiments, the protection structure further includes a second chamber inner surface proximate to the second structure end. In some embodiments, the second chamber inner surface partially bounds a second chamber interior space having a second chamber width, where the first chamber width is greater than the second chamber width. In some embodiments, the pressure hole has a third width, where the first chamber width is greater than the third width. In some embodiments, the second piston end fits in and slidably engages with the second chamber inner surface, such that the second piston outer surface is proximate to the second chamber inner surface, and such that the second piston end is in the second chamber interior space. In some embodiments, the protection structure

has a set of vent holes through the second structure end, where the set of vent holes are proximate to and outside of the second chamber inner surface.

[0103] In some embodiments, the system further includes a first track extending across a battery case proximate to a first battery case end, and a set of first end track wheels attached to a first end of the battery module enclosure, where the set of first end track wheels is configured to engage and slidably couple with the first track.

[0104] In some embodiments, the system further includes a first dividing wall bracket attached to a first end of the dividing wall, where the first dividing wall bracket is configured to engage and slidably couple with the first track.

[0105] In some embodiments, the system further includes a second track extending across the battery case proximate to a second battery case end, a set of second end track wheels attached to a second end of the battery module enclosure, where the set of second end track wheels is configured to engage and slidably couple with the second track, and a second dividing wall bracket attached to a second end of the dividing wall, where the second dividing wall bracket is configured to engage and slidably couple with the second track.

[0106] In some embodiments, each lock pawl of the set of lock pawls has a sloped shape configured to allow the first piston end to press the lock pawls into the lock pawl pockets while the piston moves in a first direction while compressing the spring, and where each lock pawl has a ratchet tooth structure configured to hinder the piston from moving in a second direction after the first piston end passes the ratchet tooth structure while the piston moves in the first direction, where the first direction is opposite the second direction.

[0107] In some embodiments, each of the lock pawls comprises a material and structure configured to elastically return to an original position partially extending outside the lock pawl pocket after being pressed into the lock pawl pocket by the first piston end, such that the ratchet tooth structure blocks a path of the first piston end from moving in the second direction after the first piston end passes the ratchet tooth structure while the piston moves in the first direction.

[0108] In some embodiments, the system is configured such that a separation air gap forms between the divider wall and the battery module enclosure when the ratchet tooth structures of the lock pawls hinder the piston from moving in the second direction, and such that vent holes fluidly communicate with a separation air gap. In some embodiments, the piston has a generally cylindrical shape with a circular cross-section shape.

[0109] A battery system embodiment includes a battery case, a first battery module enclosure, a second battery module enclosure, a divider wall, a first protection structure, and a first piston. In the embodiment, the first battery module enclosure is disposed in the battery case, where the first battery module enclosure bounds a first enclosure interior space. In the embodiment, the second battery module enclosure is disposed in the battery case. In the embodiment, the divider wall is disposed in the battery case, where the divider wall is between the first battery module enclosure and the second battery module enclosure. In the embodiment, the first protection structure has a first pressure hole. In the embodiment, the first piston is at least partially disposed in the first protection structure, where the first piston fits in, and slidably engages with, the first protection structure, where at least part of a first piston end of the first piston is in fluid communication with the first enclosure interior space via the first pressure hole, and where the first piston is configured to move, in response to a pressure above a selected threshold pressure inside the first enclosure interior space acting upon the first piston end, in a first direction in the first protection structure and to separate the first battery module enclosure from the second battery module enclosure to form a separation air gap disposed between the first battery module enclosure and the second battery module enclosure.

[0110] In some embodiments, the system further includes a first spring, and a first set of lock pawls, where the first protection structure has a first chamber inner surface proximate to a first structure end, where the first chamber inner surface at least partially bounds a first chamber interior space inside the first protection structure, where the first protection structure has the first pressure hole at the first structure end, where the first protection structure has a first vent hole through a second structure end of the first protection structure, where the first vent hole is in fluid communication with the first chamber interior space, where the first piston at least partially separates the first enclosure interior space from the first chamber interior space, where the first piston has a second piston end, where the first piston fits in, and slidably engages with, the first chamber inner surface, where the second piston end is attached to the divider wall, where the second structure end is attached to the first battery module enclosure, where the first protection structure has a first set of lock pawl pockets in the first chamber inner surface, where the first set of lock pawls are at least partially in the first set of lock pawl pockets, where the first piston includes a first piston outer surface at the first piston end, where the first piston further includes a second piston outer surface at the second piston end, where the first piston end has a first piston end width greater than a second piston end width of the second piston end, where the first piston outer surface contacts the first chamber inner surface and the first piston end is disposed in the first chamber interior space, where the second piston end is attached to the divider wall, where the first spring is disposed between the first piston end and the second structure end, where the first spring surrounds the first piston in the first chamber interior space.

[0111] In some embodiments, each of the lock pawls has a sloped shape configured to allow the first piston end to press the lock pawls into the lock pawl pockets while the first piston moves in the first direction while compressing the first spring, and each of the lock pawls has a ratchet tooth structure configured to hinder the first piston from moving in a second direction after the first piston end passes the ratchet tooth structure while the first piston moves in the first direction, where the first direction is opposite the second direction.

[0112] In some embodiments, each of the lock pawls comprises a material and structure configured to elastically return to an original position partially extending outside the respective lock pawl pocket after being pressed into the respective lock pawl pocket by the first piston end, such that the ratchet tooth structure blocks a path of the first piston end from moving in the second direction after the first piston end passes the ratchet tooth structure while the first piston moves in the first direction.

[0113] In some embodiments, the first piston has a generally cylindrical shape with a circular cross-section. In some embodiments, the first chamber interior space has a first chamber width. In some embodiments, the first protection structure further includes a second chamber inner surface proximate to the second structure end. In some embodiments, the second chamber inner surface partially bounds a second chamber interior space having a second chamber width, where the first chamber width is greater than the second chamber width. In some embodiments, where the first pressure hole has a third width, the first chamber width is greater than the third width. In some embodiments, the second piston end fits in and slidably engages with the second chamber inner surface, such that the second piston outer surface is proximate to the second chamber inner surface, and such that the second piston end is in a second chamber inner space. In some embodiments, the first protection structure has a set of vent holes through the second structure end. In some embodiments, the set of vent holes are proximate to and outside of the second chamber inner surface.

[0114] In some embodiments, the battery case has a first battery case width at a first battery case end, where the battery case has a second battery case width at a second battery case end, and where the system further includes a first track extending across the first battery case width proximate to the first battery case end, a first set of first end track wheels attached to a first end of the first battery module enclosure, where the first set of first end track wheels is engaged and slidably coupled with the first track, a second set of first end track wheels attached to a first end of the second battery module enclosure, where the second set of first end track wheels is engaged and slidably coupled with the first track, a first dividing wall bracket attached to a first end of the dividing wall, where the first dividing wall bracket is engaged and slidably coupled with the first track, a second track extending across the second battery case width proximate to the second battery case end, a first set of second end track wheels attached to a second end of the first battery module enclosure, where the first set of second end track wheels is engaged and slidably coupled with the second track, a second set of second end track wheels attached to a second end of the second battery module enclosure, where the second set of second end track wheels is engaged and slidably coupled with the second track, and a second dividing wall bracket attached to a second end of the dividing wall, where the second dividing wall bracket is engaged and slidably coupled with the second track.

[0115] A method embodiment of assembling a thermal runaway protection system for a battery includes providing a battery module enclosure including an enclosure interior space, providing a divider wall, providing a protection structure including a first chamber inner surface proximate to a first structure end, where the first chamber inner surface partially bounds a first chamber interior space, where the protection structure further includes a pressure hole at the first structure end, where the protection structure further includes a set of vent holes through a second structure end, where the vent holes are in fluid communication with the first chamber interior space, and where the protection structure further includes a set of lock pawl pockets in the first chamber inner surface, providing a piston including a first piston outer surface at a first piston end, where the piston further includes a second piston outer surface at a second piston end, where the first piston end has a first piston end width, where the second piston end has a second piston end width, where the first piston end width is greater than the second piston end width, placing a set of lock pawls in the set of lock pawl pockets, placing a spring in the first chamber interior space inside the protection structure, placing the piston into the protection structure such that the spring is surrounding the piston, such that the first piston end fits in and slidably couples with the first chamber inner surface, such that the first piston outer surface contacts the first chamber inner surface, such that the first piston end is in the first chamber interior space, such that the spring is configured to bias between the first piston end and the second structure end, and such that at least part of the first piston end is in fluid communication with the enclosure interior space via the pressure hole, attaching the second structure end to the battery module enclosure, such that the pressure hole is in fluid communication with the enclosure interior space of the battery module enclosure, and attaching the second piston end to the divider wall, such that the divider wall is proximate to the battery module enclosure.

[0116] In some embodiments, the method further includes attaching a first dividing wall bracket to a first end of the dividing wall, attaching a set of first end track wheels to a first end of the battery module enclosure, engaging and slidably coupling the first dividing wall bracket to a first track, and engaging and slidably coupling the set of first end track wheels to the first track.

[0117] In some embodiments, the method further includes attaching a second dividing wall bracket to a second end of the dividing wall, attaching a set of second end track wheels to a second end of the battery module enclosure, engaging and slidably coupling the second dividing wall bracket to a second track, and engaging and slidably coupling the set of second end track wheels to the second track.

[0118] In some embodiments, the method further includes attaching the first track to a first end of a battery case, and attaching the second track to a second end of the battery case.

[0119] While illustrative embodiments have been described with reference to illustrative drawings, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments, may be apparent to persons skilled in the pertinent art upon referencing the present disclosure. It is therefore intended that the appended claims encompass any and all of such modifications or embodiments.

Claims

1. A thermal runaway protection system for a battery, comprising: a battery module enclosure bounding an enclosure interior space; a divider wall adjacent to the battery module enclosure; a protection structure attached to the battery module enclosure, the protection structure including a first chamber inner surface proximate to a first structure end, wherein the first chamber inner surface at least partially bounds a first chamber interior space inside the protection structure, wherein the protection structure has a pressure hole at the first structure end, wherein the pressure hole is in fluid communication with the enclosure interior space of the battery module enclosure, wherein the protection structure has a vent hole through a second structure end, and wherein the vent hole is in fluid communication with the first chamber interior space; and a piston at least partially disposed in the protection structure, wherein the piston at least partially separates the enclosure interior space from the first chamber interior space, wherein the piston has a first piston end and a second piston end, wherein the piston fits in, and slidably engages with, the first chamber inner surface, wherein at least part of the first piston end is in fluid communication with the enclosure interior space via the pressure hole, and wherein the second piston end is proximate to the divider wall.

2. The system of claim 1, further comprising: a spring disposed between the first piston end and the second structure end, wherein the spring surrounds the piston in the first chamber interior space; and a set of lock pawls, wherein the protection structure further includes a set of lock pawl pockets in the first chamber inner surface, and wherein the set of lock pawls are at least partially in the set of lock pawl pockets, wherein the piston includes a first piston outer surface at the first piston end, wherein the piston further includes a second piston outer surface at the second piston end, wherein the first piston end has a first piston end width greater than a second piston end width of the second piston end, wherein the first piston end fits in, and slidably engages with, the first chamber inner surface, wherein the first piston outer surface contacts the first chamber inner surface and the first piston end is disposed in the first chamber interior space, wherein the second piston end is attached to the divider wall, and wherein the second structure end is attached to the battery module enclosure.

3. The system of claim 2, wherein the first chamber interior space has a first chamber width; wherein the protection structure further includes a second chamber inner surface proximate to the second structure end; wherein the second chamber inner surface partially bounds a second chamber interior space having a second chamber width, the first chamber width being greater than the second chamber width; wherein the pressure hole has a third width, the first chamber width being greater than the third width; wherein the second piston end fits in and slidably engages with the second chamber inner surface, such that the second piston outer surface is proximate to the second chamber inner surface, and such that the second piston end is in the second chamber interior space; wherein the protection structure has a set of vent holes through the second structure end; and wherein the set of vent holes are proximate to and outside of the second chamber inner surface.

4. The system of claim 1, further comprising: a first track extending across a battery case proximate to a first battery case end; and a set of first end track wheels attached to a first end of the battery module enclosure, the set of first end track wheels being configured to engage and slidably couple with the first track.

5. The system of claim 4, further comprising a first dividing wall bracket attached to a first end of the dividing wall, the first dividing wall bracket being configured to engage and slidably couple with the first track.

6. The system of claim 5, further comprises: a second track extending across the battery case proximate to a second battery case end; a set of second end track wheels attached to a second end of the battery module enclosure, the set of second end track wheels being configured to engage and slidably couple with the second track; and a second dividing wall bracket attached to a second end of the dividing wall, the second dividing wall bracket being configured to engage and slidably couple with the second track.

7. The system of claim 2, wherein each lock pawl of the set of lock pawls has a sloped shape configured to allow the first piston end to press the lock pawls into the lock pawl pockets while the piston moves in a first direction while compressing the spring, and wherein each lock pawl has a ratchet tooth structure configured to hinder the piston from moving in a second direction after the first piston end passes the ratchet tooth structure while the piston moves in the first direction, the first direction being opposite the second direction.

8. The system of claim 7, wherein each of the lock pawls comprises a material and structure configured to elastically return to an original position partially extending outside the lock pawl pocket after being pressed into the lock pawl pocket by the first piston end, such that the ratchet tooth structure blocks a path of the first piston end from moving in the second direction after the first piston end passes the ratchet tooth structure while the piston moves in the first direction.

9. The system of claim 8, wherein the system is configured such that a separation air gap forms between the divider wall and the battery module enclosure when the ratchet tooth structures of the lock pawls hinder the piston from moving in the second direction, and such that vent holes fluidly communicate with a separation air gap.

10. The system of claim 1, wherein the piston has a generally cylindrical shape with a circular cross-section shape.

11. A battery system comprising: a battery case; a first battery module enclosure disposed in the battery case, wherein the first battery module enclosure bounds a first enclosure interior space; a second battery module enclosure disposed in the battery case; a divider wall disposed in the battery case, the divider wall being between the first battery module enclosure and the second battery module enclosure; a first protection structure having a first pressure hole; and a first piston at least partially disposed in the first protection structure, wherein the first piston fits in, and slidably engages with, the first protection structure, wherein at least part of a first piston end of the first piston is in fluid communication with the first enclosure interior space via the first pressure hole, and wherein the first piston is configured to move, in response to a pressure above a selected threshold pressure inside the first enclosure interior space acting upon the first piston end, in a first direction in the first protection structure and to separate the first battery module enclosure from the second battery module enclosure to form a separation air gap disposed between the first battery module enclosure and the second battery module enclosure.

12. The system of claim 11, further comprising: a first spring; and a first set of lock pawls, wherein the first protection structure has a first chamber inner surface proximate to a first structure end, wherein the first chamber inner surface at least partially bounds a first chamber interior space inside the first protection structure, wherein the first protection structure has the first pressure hole at the first structure end, wherein the first protection structure has a first vent hole through a second structure end of the first protection structure, wherein the first vent hole is in fluid communication with the first chamber interior space, wherein the first piston at least partially separates the first enclosure interior space from the first chamber interior space, wherein the first piston has a second piston end, wherein the first piston fits in, and slidably engages with, the first chamber inner surface, wherein the second piston end is attached to the divider wall, wherein the second structure end is attached to the first battery module enclosure, wherein the first protection structure has a first set of lock pawl pockets in the first chamber inner surface, wherein the first set of lock pawls are at least partially in the first set of lock pawl pockets, wherein the first piston includes a first piston outer surface at the first piston end, wherein the first piston further includes a second piston outer surface at the second piston end, wherein the first piston end has a first piston end width greater than a second piston end width of the second piston end, wherein the first piston outer surface contacts the first chamber inner surface and the first piston end is disposed in the first chamber

interior space, wherein the second piston end is attached to the divider wall, wherein the first spring is disposed between the first piston end and the second structure end, wherein the first spring surrounds the first piston in the first chamber interior space.
13. The system of claim 12, wherein each of the lock pawls has a sloped shape configured to allow the first piston end to press the lock pawls into the lock pawl pockets while the first piston moves in the first direction while compressing the first spring, and each of the lock pawls has a ratchet tooth structure configured to hinder the first piston from moving in a second direction after the first piston end passes the ratchet tooth structure while the first piston moves in the first direction, the first direction being opposite the second direction.
14. The system of claim 13, wherein each of the lock pawls comprises a material and structure configured to elastically return to an original position partially extending outside the respective lock pawl pocket after being pressed into the respective lock pawl pocket by the first piston end, such that the ratchet tooth structure blocks a path of the first piston end from moving in the second direction after the first piston end passes the ratchet tooth structure while the first piston moves in the first direction.
15. The system of claim 12, wherein the first piston has a generally cylindrical shape with a circular cross-section; wherein the first chamber interior space has a first chamber width; wherein the first protection structure further includes a second chamber inner surface proximate to the second structure end; wherein the second chamber inner surface partially bounds a second chamber interior space having a second chamber width, the first chamber width being greater than the second chamber width; wherein the first pressure hole has a third width, the first chamber width being greater than the third width; wherein the second piston end fits in and slidably engages with the second chamber inner surface, such that the second piston outer surface is proximate to the second chamber inner surface, and such that the second piston end is in a second chamber inner space; wherein the first protection structure has a set of vent holes through the second structure end; and wherein the set of vent holes are proximate to and outside of the second chamber inner surface.
16. The system of claim 11, wherein the battery case has a first battery case width at a first battery case end, wherein the battery case has a second battery case width at a second battery case end, and wherein the system further comprises: a first track extending across the first battery case width proximate to the first battery case end; a first set of first end track wheels attached to a first end of the first battery module enclosure, the first set of first end track wheels being engaged and slidably coupled with the first track; a second set of first end track wheels attached to a first end of the second battery module enclosure, the second set of first end track wheels being engaged and slidably coupled with the first track; a first dividing wall bracket attached to a first end of the dividing wall, the first dividing wall bracket being engaged and slidably coupled with the first track; a second track extending across the second battery case width proximate to the second battery case end; a first set of second end track wheels attached to a second end of the first battery module enclosure, the first set of second end track wheels being engaged and slidably coupled with the second track; a second set of second end track wheels attached to a second end of the second battery module enclosure, the second set of second end track wheels being engaged and slidably coupled with the second track; and a second dividing wall bracket attached to a second end of the dividing wall, the second dividing wall bracket being engaged and slidably coupled with the second track.
17. A method of assembling a thermal runaway protection system for a battery, comprising: providing a battery module enclosure including an enclosure interior space; providing a divider wall; providing a protection structure including a first chamber inner surface proximate to a first structure end, wherein the first chamber inner surface partially bounds a first chamber interior space, wherein the protection structure further includes a pressure hole at the first structure end, wherein the protection structure further includes a set of vent holes through a second structure end,

wherein the vent holes are in fluid communication with the first chamber interior space, and wherein the protection structure further includes a set of lock pawl pockets in the first chamber inner surface; providing a piston including a first piston outer surface at a first piston end, wherein the piston further includes a second piston outer surface at a second piston end, wherein the first piston end has a first piston end width, wherein the second piston end has a second piston end width, wherein the first piston end width is greater than the second piston end width; placing a set of lock pawls in the set of lock pawl pockets; placing a spring in the first chamber interior space inside the protection structure; placing the piston into the protection structure such that the spring is surrounding the piston, such that the first piston end fits in and slidably couples with the first chamber inner surface, such that the first piston outer surface contacts the first chamber inner surface, such that the first piston end is in the first chamber interior space, such that the spring is configured to bias between the first piston end and the second structure end, and such that at least part of the first piston end is in fluid communication with the enclosure interior space via the pressure hole; attaching the second structure end to the battery module enclosure, such that the pressure hole is in fluid communication with the enclosure interior space of the battery module enclosure; and attaching the second piston end to the divider wall, such that the divider wall is proximate to the battery module enclosure.

18. The method of claim 17, further comprising: attaching a first dividing wall bracket to a first end of the dividing wall; attaching a set of first end track wheels to a first end of the battery module enclosure; engaging and slidably coupling the first dividing wall bracket to a first track; and engaging and slidably coupling the set of first end track wheels to the first track.

19. The method of claim 18, further comprising: attaching a second dividing wall bracket to a second end of the dividing wall; attaching a set of second end track wheels to a second end of the battery module enclosure; engaging and slidably coupling the second dividing wall bracket to a second track; and engaging and slidably coupling the set of second end track wheels to the second track.

20. The method of claim 19, further comprising: attaching the first track to a first end of a battery case; and attaching the second track to a second end of the battery case.